

SESSION 19

Eigenvalues & Eigenvectors

Eigen Equation $Av = \lambda v$ · Characteristic Equation · Geometric Interpretation · Properties

The Eigenvalue Equation ($Av = \lambda v$)

The Characteristic Equation

Finding Eigenvalues and Eigenvectors

Geometric Intuition of Eigenvalues

Algebraic vs Geometric Multiplicity

Properties of Eigenvalues

Eigenvalues of Special Matrices

Python Calculation of Eigenvalues

Session Notes — Complete Reference

Playlist: Maths for Machine Learning · Instructor: Nitish Singh (CampusX)



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01 • The Eigenvalue Equation ($\mathbf{Av} = \lambda \mathbf{v}$)

An **eigenvector** of a square matrix $\mathbf{A}$ is a non-zero vector $\mathbf{v}$ that does not change direction when multiplied by $\mathbf{A}$. It is only scaled by a factor called the **eigenvalue** (λ).

EIGEN EQUATION

$$\mathbf{A} \cdot \mathbf{v} = \lambda \cdot \mathbf{v}$$

02 • The Characteristic Equation

To find eigenvalues, we rewrite the equation as $(A - \lambda I)\mathbf{v} = \mathbf{0}$. Since $\mathbf{v}$ is non-zero, the matrix $(A - \lambda I)$ must be singular:

CHARACTERISTIC EQUATION

$$\det(A - \lambda I) = 0$$

Solving this polynomial equation yields the eigenvalues λ .

03 · Finding Eigenvalues and Eigenvectors

Once we solve for λ , we plug each eigenvalue back into $(A - \lambda I)\mathbf{v} = \mathbf{0}$ and solve for the corresponding eigenvector $\mathbf{v}$ (which lies in the null space of $A - \lambda I$).

04 · Geometric Intuition of Eigenvalues

Linear transformations typically rotate and scale vectors. Eigenvectors are the **axes of the transformation** — vectors along these axes do not rotate, they only stretch or shrink by the factor λ .

05 · Algebraic vs Geometric Multiplicity

- **Algebraic Multiplicity:** How many times an eigenvalue appears as a root of the characteristic polynomial.
- **Geometric Multiplicity:** The number of linearly independent eigenvectors associated with that eigenvalue.

06 · Properties of Eigenvalues

For any $n \times n$ matrix A :

- **Sum of Eigenvalues:** Equals the **trace** of the matrix (sum of diagonal entries).
- **Product of Eigenvalues:** Equals the **determinant** of the matrix.

07 · Eigenvalues of Special Matrices

- **Symmetric Matrices:** Always have real eigenvalues, and their eigenvectors are orthogonal.
- **Triangular Matrices:** Eigenvalues are simply the entries on the main diagonal.

08 · Python Calculation of Eigenvalues

```
import numpy as np

A = np.array([[2, 1], [1, 2]])
eigenvalues, eigenvectors = np.linalg.eig(A)

print("Eigenvalues:", eigenvalues)
print("Eigenvectors:
", eigenvectors)
```

09 · Quick Reference Card



Eigenvalues Cheat Sheet

- **Equation:** $A\mathbf{v} = \lambda\mathbf{v}$.
- **Trace:** $\text{Tr}(A) = \sum \lambda_i$.
- **Determinant:** $\det(A) = \prod \lambda_i$.